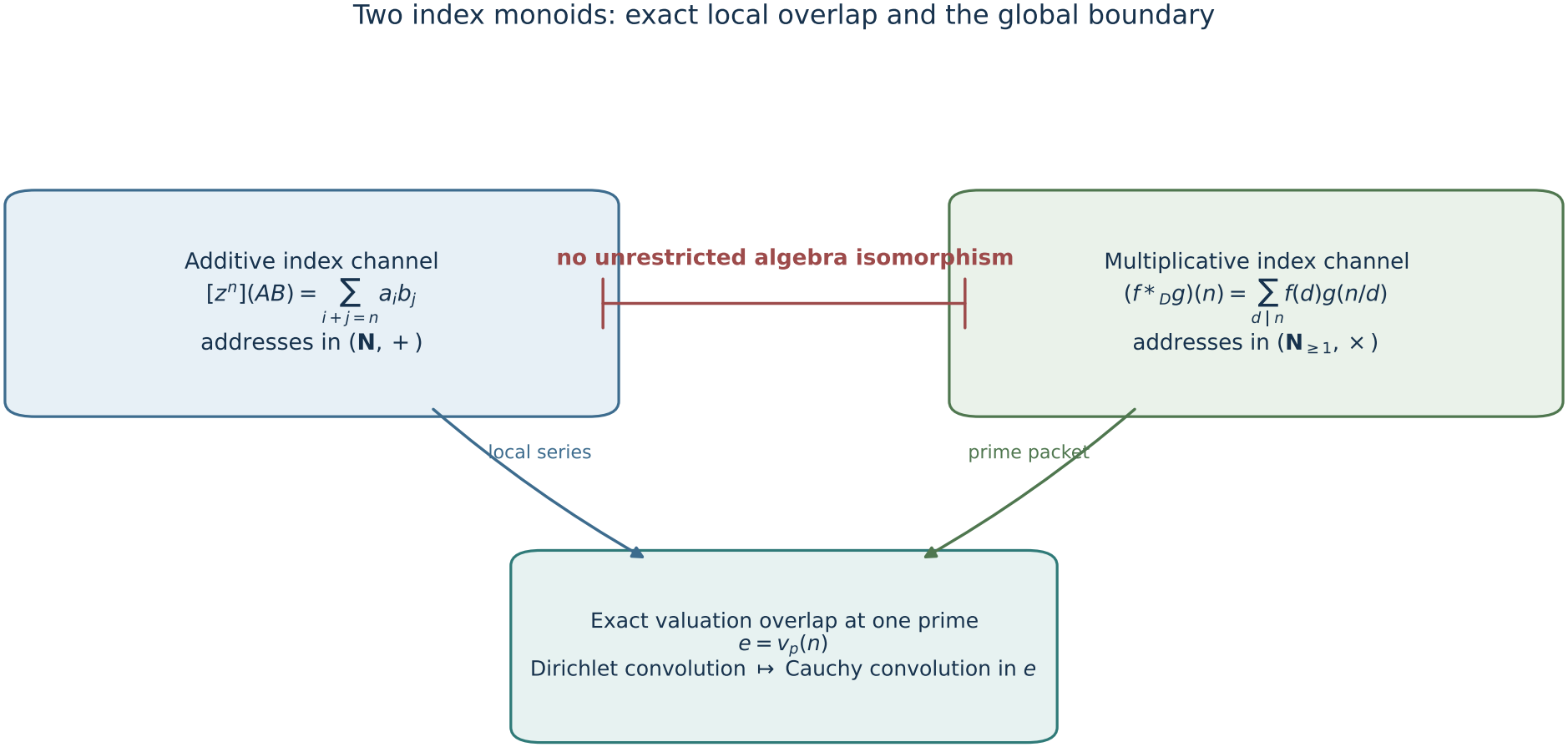
Two index monoids: exact local overlap and the global boundary
Additive index channel
$[z^n](AB) = \sum_{i+j=n} a_i b_j$
addresses in $(\mathbf{N}, +)$
no unrestricted algebra isomorphism
Multiplicative index channel
$(f *_D g)(n) = \sum_{d \mid n} f(d) g(n/d)$
addresses in $(\mathbf{N}_{\geq 1}, \times)$
local series
prime packet
Exact valuation overlap at one prime
$e = v_p(n)$
Dirichlet convolution $\mapsto$ Cauchy convolution in e
A valid bridge must name its kernel, transform, essential image, and the information it forgets.